9th Class • Punjab Board • PCTB New Syllabus

Chapter 10: Reproduction in Plants

Solved notes in book order — MCQs, short questions, long questions, inquisitive questions and key points.

Chapter 10 • Units 10.1 – 10.3 • Exercise solved in book order

What this chapter is about

• Reproduction produces new organisms of the same kind. Asexual reproduction does not involve gamete fusion; sexual reproduction involves male and female gametes.
• This chapter covers binary fission, budding, spores, natural vegetative structures, cutting, grafting and the advantages and disadvantages of vegetative propagation.
• It then explains flower structure, alternation of generations, pollen and embryo-sac formation, pollination, double fertilization, seeds and fruits.
• The complete exercise is solved in book order, including all nine comparison parts and a labelled flower diagram.

A. Multiple Choice Questions

Select the correct option. A short reason follows each answer.

1. Which of the following organisms commonly reproduce by binary fission?

a) Yeast
b) Bacteria
c) Rhizopus
d) Plants

Answer: b) Bacteria — A bacterial cell replicates its DNA and divides into two genetically identical daughter cells by binary fission.

2. What is the primary method of reproduction in yeast?

a) Binary fission
b) Spore formation
c) Budding
d) Fragmentation

Answer: c) Budding — Yeast commonly forms a small outgrowth called a bud, which grows into a new yeast cell.

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3. Which of the following statements is true about spore formation in fungi?

a) They produce spores during sexual reproduction
b) They produce two kinds of spores
c) Spores can only grow into new fungi in dry environments
d) Spores are produced to withstand harsh conditions

Answer: d) Spores are produced to withstand harsh conditions — Thick-walled fungal spores survive unfavourable conditions and germinate when conditions become suitable.

4. What happens in some bacteria during harsh conditions?

a) Creation of a bud that detaches from the cell
b) Formation of thick-walled endospores
c) Splitting the cell into two identical daughter cells
d) Fusion of two bacterial cells

Answer: b) Formation of thick-walled endospores — An endospore protects the bacterial genetic material until favourable conditions return.

5. Which of the following is an example of vegetative propagation through runners?

a) Potato
b) Strawberry
c) Onion
d) Ginger

Answer: b) Strawberry — Strawberry produces above-ground runners (stolons), and new plants form at their nodes.

6. Which plant propagates through tubers?

a) Onion
b) Potato
c) Ginger
d) Garlic

Answer: b) Potato — A potato tuber is an underground fleshy stem whose eyes are buds that can produce new plants.

7. The horizontal aboveground stem, which produces leaves and roots at its nodes;

a) Stolon
b) Bulb
c) Rhizome
d) Corm

Answer: a) Stolon — A stolon or runner grows horizontally above the ground; a rhizome grows below ground.

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8. Which of these does NOT help a plant for vegetative propagation?

a) Rhizome
b) Corm
c) Runner
d) Flower

Answer: d) Flower — A flower is the sexual reproductive structure; rhizomes, corms and runners are vegetative structures.

9. Which part of the flower is responsible for producing pollen?

a) Stigma
b) Anther
c) Ovary
d) Petal

Answer: b) Anther — Pollen sacs inside the anther produce microspores, which develop into pollen grains.

10. Which of the following is NOT a part of carpel?

a) Filament
b) Style
c) Stigma
d) Ovary

Answer: a) Filament — A carpel consists of stigma, style and ovary; the filament supports the anther in a stamen.

11. Which structure forms the female gametophyte in flowering plants?

a) Pollen grain
b) Ovule
c) Anther
d) Sepal

Answer: b) Ovule — The female gametophyte (embryo sac) develops inside the ovule from a functional megaspore.

12. The male gametophyte in flowering plants is known as:

a) Pollen grain
b) Embryo sac
c) Ovary
d) Carpel

Answer: a) Pollen grain — A microspore develops by mitosis into a pollen grain, the male gametophyte.

13. In the life cycle of flowering plants, which structure is triploid (3n)?

a) Egg
b) Fusion nucleus
c) Endosperm nucleus
d) Sperm

Answer: c) Endosperm nucleus — One haploid sperm fuses with the diploid fusion nucleus, producing a triploid endosperm nucleus.

14. Embryo sac is formed inside;

a) Filament
b) Anther
c) Style
d) Ovule

Answer: d) Ovule — The ovule acts as the megasporangium and contains the developing embryo sac.

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15. Double fertilization involves;

a) Fertilization of the egg by two male gametes
b) Fertilization of two eggs in the same embryo sac by two sperms
c) Fertilization of the egg and the fusion nucleus by two sperms
d) Fertilization of the egg and the tube cell by two sperms

Answer: c) Fertilization of the egg and the fusion nucleus by two sperms — One sperm forms the zygote with the egg; the other forms the triploid endosperm nucleus with the fusion nucleus.

B. Write Short Answers

1. Write a short note on budding in yeast.

Budding is asexual reproduction in which a small outgrowth, called a **bud**, forms on the parent yeast cell. The nucleus divides and one daughter nucleus enters the bud. The bud grows and may separate as a new yeast cell, or remain attached and form a chain of cells. The new cells are genetically identical to the parent.

2. Write a short note on spore formation in fungi.

Spores are thick-walled asexual reproductive cells. Fungi such as *Rhizopus* (bread mould) produce many spores inside sac-like **sporangia**. When mature, the sporangium bursts and releases the spores. They tolerate unfavourable conditions and germinate into new fungi when moisture, food and other suitable conditions are available.

3. What are the advantages of spore formation in fungi and bacteria?

In fungi, spores are produced in large numbers, are easily dispersed and survive harsh conditions because of their thick walls; they later germinate into new fungi. In bacteria, thick-walled **endospores** protect the cell's genetic material during heat, dryness and food shortage. A bacterial endospore is mainly a survival structure: the bacterium becomes active again when conditions improve.

4. Describe how vegetative propagation occurs through runners.

A runner or **stolon** is a horizontal stem that grows above the ground. At its nodes, leaves grow upwards and roots grow downwards into the soil. These nodes develop into new plants, which can live independently when the runner dries or is separated. Strawberry reproduces naturally in this way.

5. State how potatoes reproduce through tubers.

A potato tuber is a fleshy underground stem that stores food. Its small depressions, called **eyes**, are actually buds. When a tuber or a piece containing an eye is planted, the bud uses the stored food to produce a shoot and roots. It grows into a new potato plant genetically identical to the parent.

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6. Describe the advantages and disadvantages of vegetative propagation.

Advantages: It produces many plants in a short time, does not need seeds or pollinators, and preserves useful characters such as good fruit quality because offspring are identical to the parent. **Disadvantages:** It produces little or no genetic variation, so all plants may be equally sensitive to the same disease, pest or environmental change. Overcrowding can also occur when new plants form close to the parent.

7. Name the four whorls present in a flower and also tell the components of each whorl.

Floral whorl	Components and function
1. Calyx	The outermost whorl, made of green leaf-like **sepals**. It protects the developing flower bud.
2. Corolla	The second whorl, made of **petals**. Coloured petals attract pollinators.
3. Androecium	The male whorl, made of **stamens**. Each stamen has an anther that produces pollen and a supporting filament.
4. Gynoecium	The innermost female whorl, made of **carpels**. Each carpel has stigma, style and ovary; the ovary contains ovules.

8. Briefly describe the formation of egg cell and polar nuclei within embryo sac of a flower.

The ovule contains a diploid megaspore mother cell. It undergoes **meiosis** and forms four haploid megaspores, but only one remains functional. This megaspore undergoes three mitotic divisions and produces eight haploid nuclei inside the embryo sac.

One nucleus becomes part of the **egg cell**, the female gamete. Two nuclei move to the centre as **polar nuclei**; they fuse to form the diploid **fusion nucleus**. During double fertilization, a sperm fuses with the egg and the other sperm fuses with this fusion nucleus.

9. Differentiate between:

i. Asexual and sexual reproduction

Asexual reproduction	Sexual reproduction
Does not involve the fusion of gametes.	Involves fusion of male and female gametes.
Usually involves one parent.	Involves gametes from one or two parents.

Asexual reproduction	Sexual reproduction
Offspring are usually genetically identical to the parent.	Offspring show genetic variation.
Usually fast and produces many offspring.	Usually slower and involves flowers, spores, gametes and fertilization in plants.
Examples: binary fission, budding and vegetative propagation.	Example: seed formation after double fertilization in flowering plants.

ii. Binary fission in bacteria and amoeba

Binary fission in bacteria	Binary fission in amoeba
The single circular DNA molecule replicates; there is no mitotic spindle.	The nucleus first divides by mitosis.
Daughter DNA molecules move to opposite sides.	Two daughter nuclei form before cytoplasmic division.
Cell membrane pinches in and a new cross-wall forms in the middle.	The flexible cell membrane constricts and cytoplasm divides.
Two identical daughter bacteria form.	Two identical daughter amoebae form.

iii. Stolon and rhizome

Stolon (runner)	Rhizome
A horizontal stem that grows **above the ground**.	A horizontal stem that grows **below the ground**.
Roots and leaves form at nodes and produce new plants.	New roots and shoots also form at nodes.
Usually slender and spreads over the soil surface.	Usually thicker and stores food underground.
Example: strawberry.	Examples: ginger, ferns and sugarcane.

iv. Bulb and corm

Bulb	Corm
Has a very short underground stem with a bud and many **fleshy leaves**.	Consists mostly of a swollen underground **stem**, with thin brown non-functional leaves outside.
Food is stored mainly in the fleshy leaf bases.	Food is stored mainly in the stem tissue.

Bulb	Corm
Examples: onion, tulip and lily.	Examples given in the textbook: dasheen and garlic.

v. Cutting and grafting

Cutting	Grafting
A piece of stem or root is cut and grown into a complete new plant.	A bud or small stem of one plant is joined to the stem or root system of another.
Only one parent plant is needed.	Parts of two plants are joined; the upper part is the scion and the rooted part is the stock.
The cutting develops its own new roots and shoots.	The joined parts grow as one plant and combine useful characters.
Examples: rose, grapevine and sweet potato.	Common in commercial fruit and ornamental trees.

vi. Vegetative propagation and artificial propagation

Natural vegetative propagation	Artificial propagation
Occurs naturally without human help.	Carried out by humans using vegetative parts.
Uses natural structures such as runners, tubers, rhizomes, bulbs, corms, suckers or modified leaves.	Uses methods such as cutting, grafting, layering and tissue culture.
New plants develop at natural nodes or buds.	A grower selects and treats the plant part to obtain desired new plants.
Example: strawberry plants forming on runners.	Example: rose plants grown from stem cuttings.

vii. Male and female gametophytes

Male gametophyte	Female gametophyte
The **pollen grain**.	The **embryo sac**.
Develops from a microspore inside the anther's pollen sac.	Develops from a functional megaspore inside the ovule.
Has a tube cell and a generative cell; the generative cell forms two sperms.	Contains the egg cell and the central cell with two polar nuclei, among other cells.
Produces male gametes (sperms).	Produces the female gamete (egg).

viii. Calyx and corolla

Calyx	Corolla
The outermost floral whorl.	The second floral whorl, inside the calyx.
Made of sepals.	Made of petals.
Usually green and leaf-like.	Usually coloured and conspicuous.
Protects the inner parts of a developing bud.	Attracts insects and other pollinators.

ix. Stamen and carpel

Stamen	Carpel
Male reproductive structure of a flower.	Female reproductive structure of a flower.
Part of the androecium.	Part of the gynoecium.
Consists of anther and filament.	Consists of stigma, style and ovary.
Anther contains pollen sacs and produces microspores/pollen.	Ovary contains ovules; stigma receives pollen and style supports the stigma.

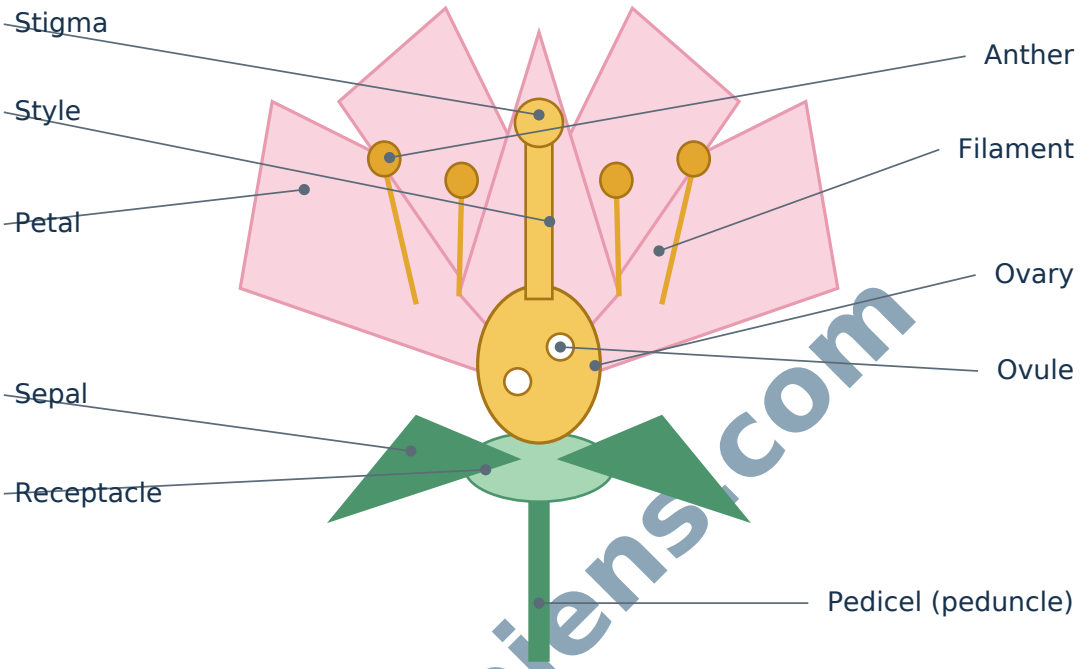
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10. Label the given diagram of flower.

The main labels are **stigma, style, anther, filament, petal, ovary, ovule, sepal, receptacle and pedicel (peduncle)**. Their positions are shown below.

Answer diagram — parts of a typical flower

Labelled Diagram of a Flower



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C. Write Answers in Detail

1. Explain the process of binary fission in bacteria and describe how it leads to the formation of two daughter bacteria.

Binary fission is a simple asexual division in which one bacterial cell forms two genetically identical daughter cells. It occurs in favourable conditions when the cell has grown sufficiently.

1. DNA replication: The bacterial DNA makes an exact copy of itself, so each future daughter cell will receive the same genetic information.

2. Separation of DNA: The two daughter DNA molecules move towards opposite sides of the elongating cell.

3. Inward pinching: The cell membrane begins to pinch inwards at the middle of the cell. The cytoplasm is divided into two nearly equal portions.

4. Cross-wall formation: A new cell wall forms across the middle and completes the separation.

5. Daughter cells: Two independent daughter bacteria are produced. Each contains one copy of the original DNA and grows to the size of the parent before dividing again.

One parent bacterium → DNA replication → inward pinching + cross-wall → two identical daughter bacteria

2. What do you mean by vegetative propagation? Differentiate among different plant structures modified for vegetative propagation.

Vegetative propagation is asexual reproduction in which a new plant develops from a **root, stem or leaf** of the parent plant. It may be natural or artificial. The offspring are genetically identical to the parent and usually grow quickly.

Natural vegetative structures

Modified structure	Main identifying feature and method	Example
Stolon / runner	Horizontal **above-ground** stem; roots and shoots form at nodes.	Strawberry
Tuber	Fleshy **underground stem** with buds called eyes; each eye can grow into a plant.	Potato
Rhizome	Horizontal **underground stem** with nodes that produce roots and shoots.	Ginger, fern, sugarcane
Bulb	Very short underground stem with a bud and food-storing **fleshy leaves**.	Onion, tulip, lily

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Modified structure	Main identifying feature and method	Example
Corm	Swollen underground stem with no fleshy storage leaves; thin brown leaves remain outside.	Dasheen, garlic (book examples)
Sucker	New shoot from the base or underground root of the parent; becomes independent after forming roots.	Banana, raspberry
Modified leaf	Buds at the leaf margins grow into new plants when the leaf touches the ground.	Bryophyllum

3. Describe the ways by which humans can grow new plants by using the vegetative parts of the parent plants?

Growing new plants from vegetative parts by human methods is called **artificial propagation**. It is used to multiply plants with desirable characters and to increase crop production.

1. Stem cutting: A healthy piece of stem containing nodes is cut from the parent and planted in suitable moist soil or another rooting medium. It forms adventitious roots and new shoots. Roses, ornamental shrubs and grapevines are commonly grown this way.

2. Root cutting: A piece of root is cut and planted under suitable conditions. It produces new shoots and roots, forming a complete plant. Sweet potato is an example.

3. Grafting: A bud or small stem (**scion**) of a plant with desired characters is attached to the rooted stem (**stock**) of another plant of the same type. Their tissues unite and grow as one plant. Grafting combines useful qualities of both plants and is widely used for commercial fruit and ornamental trees.

4. Other methods: In **layering**, a branch is encouraged to form roots while still attached to the parent and is later separated. In **tissue culture**, many plantlets are grown from a small piece of plant tissue under sterile, controlled conditions. These methods are useful when rapid multiplication is required.

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4. Define sporophyte and gametophyte. State their roles in the life cycle of plants.

Sporophyte generation	Gametophyte generation
The **diploid (2n)** generation of a plant.	The **haploid (1n)** generation of a plant.
Produces spores by **meiosis**.	Produces gametes by **mitosis**.
Spores are haploid and germinate to form gametophytes.	Haploid male and female gametes fuse during fertilization.
The familiar flowering plant is the sporophyte.	In flowering plants, pollen grain and embryo sac are the very small gametophytes.
A diploid zygote develops into the next sporophyte.	Its gametes form the diploid zygote, restoring the sporophyte generation.

These two generations come one after the other. This regular alternation between a spore-producing sporophyte and a gamete-producing gametophyte is called **alternation of generations**.

5. Explain the lifecycle of flowering plants, focusing on the alternation between the gametophyte and sporophyte generations.

1. Sporophyte plant: The familiar diploid (2n) flowering plant bears flowers. Its anthers and ovules contain diploid spore mother cells.

2. Formation of spores: In the anther, each microspore mother cell undergoes meiosis and forms haploid microspores. In the ovule, the megaspore mother cell undergoes meiosis and forms four haploid megaspores; only one remains functional.

3. Gametophyte generation: A microspore undergoes mitosis and becomes the **pollen grain** (male gametophyte) with a tube cell and a generative cell. The functional megaspore undergoes mitosis and forms the **embryo sac** (female gametophyte), containing an egg and polar nuclei.

4. Pollination: Pollen grains are transferred from anther to stigma. The tube cell forms a pollen tube, which grows through the style towards the ovule. The generative cell forms two haploid sperms.

5. Double fertilization: The pollen tube enters the ovule through the micropyle. One sperm fuses with the egg and forms a **diploid zygote (2n)**. The other fuses with the diploid fusion nucleus and forms a **triploid endosperm nucleus (3n)**.

6. Return to sporophyte: The zygote divides by mitosis and forms the embryo; the endosperm nucleus forms nourishing endosperm. The ovule becomes a seed and the ovary becomes a fruit. When the seed germinates, the embryo grows into a new diploid sporophyte plant, completing the cycle.

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Sporophyte (2n) → meiosis → spores (n) → gametophytes (n) → gametes (n) → fertilization → zygote (2n) → new sporophyte

6. Describe how the female gametophyte (embryo sac) develops within the ovule of a flower.

1. Megaspore mother cell: The ovule acts as the megasporangium and contains one diploid (2n) megaspore mother cell.

2. Meiosis: The megaspore mother cell divides by meiosis and forms four haploid (n) megaspores. Three degenerate and only one remains functional.

3. Three mitotic divisions: The nucleus of the functional megaspore divides three times by mitosis without complete cell division after each division. Thus, eight haploid nuclei are formed inside the developing embryo sac.

4. Central cell: One nucleus from each end moves to the centre. These two **polar nuclei** fuse and form a diploid **fusion nucleus** inside the large central cell.

5. Egg apparatus: At the micropylar end, three cells organise as one **egg cell** and two supporting synergids. The egg is the female gamete.

6. Antipodal end: The three cells at the opposite end are called antipodals. The mature embryo sac is therefore generally **seven-celled but eight-nucleate**: egg cell, two synergids, three antipodals and one central cell containing the fusion nucleus.

The mature embryo sac is the female gametophyte. After pollen arrives, one sperm fertilizes the egg and the other fuses with the fusion nucleus during double fertilization.

D. Inquisitive Questions

1. Why are spores considered an adaptation for survival in harsh environmental conditions?

Spores have thick protective walls and very low metabolic activity. They can tolerate dryness, food shortage and other unfavourable conditions that would damage an actively growing organism. They remain dormant until moisture, food and a suitable temperature return, and then germinate.

Fungal spores are also very small and produced in large numbers, so air can disperse them to new habitats. Bacterial endospores mainly protect the original cell rather than increase its number. In both cases, the adaptation prevents the loss of the organism's genetic line during harsh periods.

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2. How do asexual and sexual reproduction contribute differently to genetic diversity of plant populations?

Asexual reproduction produces offspring from one parent without gamete fusion. The offspring are usually genetically identical, so it adds little or no new genetic diversity. It preserves successful characters and multiplies plants rapidly, but a whole population may be vulnerable to the same disease or environmental change.

Sexual reproduction creates variation through meiosis and the fusion of gametes. New combinations of alleles appear in the offspring, so individuals differ from one another and from their parents. This diversity gives some individuals a better chance to survive diseases and changing conditions and provides material for adaptation and evolution.

3. How does the pollen tube facilitate the process of fertilization in flowering plants?

After a compatible pollen grain reaches the stigma, its tube cell forms a pollen tube. The tube grows down through the stigma and style and guides the two male gametes safely towards the ovary. It enters the ovule through the micropyle and reaches the embryo sac.

The generative cell forms two sperms, which are delivered by this tube. One sperm can then fuse with the egg to form the zygote, and the other with the fusion nucleus to form the triploid endosperm nucleus. Thus, the pollen tube makes double fertilization possible without the sperms having to swim through outside water.

E. More Important Questions (from the text)

1. Differentiate between self-pollination and cross-pollination.

Self-pollination	Cross-pollination
Pollen moves from anther to stigma of the same flower or another flower on the **same plant**.	Pollen moves from the anther of one plant to the stigma of a flower on a **different plant of the same species**.
Usually produces less genetic variation.	Usually produces more genetic variation.
Does not always require an external pollinating agent.	Often needs wind, water, insects or other animals to carry pollen.
Preserves parental characters more reliably.	Combines characters from different parents.

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2. What is double fertilization? Why is it important?

Double fertilization is a special feature of flowering plants in which two fusions occur in the same embryo sac. One sperm fuses with the haploid egg to form the **diploid zygote (2n)**, which develops into the embryo. The second sperm fuses with the diploid fusion nucleus to form the **triploid endosperm nucleus (3n)**, which develops into endosperm and nourishes the embryo. It therefore forms both the future plant and its food supply.

3. What is parthenocarpy? Give examples.

Parthenocarpy is the development of a fruit **without fertilization**. Because fertilization does not occur, seeds are not formed. Bananas and seedless varieties of grapes are examples of parthenocarpic fruits.

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Points to Remember

Asexual reproduction: no gamete fusion; offspring are usually identical. **Sexual reproduction:** gametes fuse and offspring show variation.

Binary fission: bacterial DNA replicates, daughter DNA separates, membrane pinches in and a cross-wall forms two daughter cells. In amoeba, the nucleus divides before the cytoplasm.

Budding: a yeast bud grows from the parent and becomes a new cell. **Spores:** thick-walled cells that survive harsh conditions and germinate when suitable.

Natural vegetative structures: strawberry — stolon; potato — tuber; ginger — rhizome; onion — bulb; dasheen/garlic — corm (book examples); banana — sucker; Bryophyllum — leaf-margin buds.

Artificial propagation: cuttings and grafting are the two main methods described in the book; layering and tissue culture are also vegetative methods.

Four floral whorls: calyx (sepals), corolla (petals), androecium (stamens) and gynoecium (carpels).

Stamen: anther + filament. **Carpel:** stigma + style + ovary. The anther produces pollen; the ovule contains the embryo sac.

Alternation of generations: diploid sporophyte makes haploid spores by meiosis; haploid gametophyte makes gametes by mitosis; fertilization restores diploid sporophyte.

Male gametophyte: pollen grain with tube cell and generative cell. **Female gametophyte:** embryo sac formed from one functional megaspore.

Mature embryo sac: usually seven cells and eight nuclei, with one egg, two synergids, three antipodals and a central cell containing the fusion nucleus.

Double fertilization: sperm + egg → zygote (2n); sperm + fusion nucleus (2n) → endosperm nucleus (3n).

After fertilization: zygote → embryo; endosperm nucleus → endosperm; ovule → seed; ovary → fruit. Seed germination produces the next sporophyte.

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How to attempt this chapter in the paper

- Learn vegetative structures with one identifying feature and one example. 'Above ground' identifies a stolon; 'eyes' identify a tuber; 'fleshy leaves' identify a bulb.
- Use 2n for sporophyte and fusion nucleus, n for spores, gametophytes and gametes, and 3n for endosperm nucleus.
- In life-cycle answers, follow the order: spore mother cells → meiosis → spores → gametophytes → gametes → pollination → double fertilization → seed and fruit → new sporophyte.
- Draw and label the flower neatly: stigma, style, ovary, ovule, anther, filament, petal, sepal, receptacle and pedicel.